Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Data Engineer and Software Engineer with 8+ years building scalable data pipelines, ETL/ELT systems, and cloud-native data infrastructure at scale. At Entrata (PropTech SaaS, 100K+ MAU), designed and operated high-volume data ingestion, processing, and distribution pipelines across 1.6M+ financial records on SQL and NoSQL systems on AWS - automated data quality gates, compliance enforcement, 30% query performance improvement across 200+ production queries, and operational reliability sustained at 99.5% uptime. Built PHI-compliant data pipelines on AWS S3 with zero security incidents. Eliminated 80% of manual operational workload via Python/Bash orchestration automation. Also built AI/ML data workflows: LeaseI evaluation harness (225-query benchmark dataset, LLM-as-judge scoring, regression gating across 1,240 lease PDFs) and AgentProbe (multi-run AI evaluation benchmarking infrastructure). M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

Evaluation infrastructure: github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

Data Pipelines & ETL/ELT | Large-Scale Data Ingestion | Apache Spark / PySpark | Distributed Data Processing | AWS Data Services (S3, EC2, RDS, Glue) | Python | SQL / PL/SQL | PostgreSQL · MySQL · Oracle | NoSQL / Non-Relational Databases | Data Partitioning & Query Optimization | Data Quality & Integrity Monitoring | Pipeline Automation, Orchestration & Monitoring | AI/ML Data Workflows | LLM Evaluation Infrastructure | Scalable Data Architectures | Cloud-Native Data Infrastructure | Schema Design & Data Modeling

WORK EXPERIENCE

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer - ETL/ELT Pipelines · Data Infrastructure · AWS · SQL Optimization · Compliance

- **Designed and operated scalable ETL/ELT data pipelines** for high-volume data ingestion, processing, and transformation of 1.6M+ HUD and Tax Credit financial records across five years - parameterized SQL import scripts with automated data quality gates (pre-insertion validation, AMI threshold logic, eligibility tier enforcement) across 100K+ unit portfolios; zero pipeline failures in production.
- **Optimized 200+ SQL queries** across all client environments - execution plan tuning, indexing strategies, database partitioning, caching, and audit log tables - **30% performance improvement** sustained across high-traffic production workloads; led database architecture decisions with data architects for complex production schemas (PostgreSQL, MySQL, Oracle, PL/SQL).
- **Built PHI-compliant data pipeline on AWS S3** with Libsodium encryption - encrypted 100% of resident sensitive data (SSNs, phone numbers); updated 50%+ of the product codebase for secure data insertion and retrieval; **zero security incidents; 100% PHI compliance** across all client environments.
- **Eliminated 80% of manual operational workload** by designing scalable Python and Bash data orchestration automation pipelines with idempotent steps, validation gates, operational monitoring, and reliability checks across each downstream stage - sustained compliance and operational reliability with zero violations across 7+ years.
- **Led cloud data migration** of thousands of client documents from legacy server storage to AWS S3 - database data creation, physical file migration, schema redesign migrating hundreds of thousands of records to a modern structure; resolved 90% of storage constraints; **zero data loss**.
- **Automated income eligibility and compliance data workflows** for Section 8/HCV, LIHTC, and HUD portfolios - AMI threshold calculations, recertification pipelines, and household data validation across regulated financial data; zero compliance violations.
- Shipped 20+ full-stack production data modules (PHP, Python, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints for 100K+ monthly active users - **99.5% uptime** sustained over 6+ consecutive months; managed CRM data pipelines (Salesforce, Clay, n8n) connecting product data to commercial systems.

Entrata India (Db Xento Systems)

Jan 2017 - Sep 2018

Associate / Trainee Software Engineer - Full-Stack · Data Pipelines · Schema Design · AWS S3

- Built data ingestion and transformation modules in PHP and Python - SQL import scripts for 1.6M financial records, data normalization, schema design for affordable housing compliance workflows.
- Led cloud migration of thousands of client documents to AWS S3; redesigned outdated document storage database architecture migrating hundreds of thousands of records to a modern schema; delivered 6 months of post-deployment stabilization support.

George Mason University - Research Associate

Aug 2024 - Aug 2025

Research Associate - Data Pipelines · Statistical Modeling · ML · Anomaly Detection

- Built multi-source data collection, transformation, and validation pipelines (Python, R) on WHO and UK Road Safety datasets across 190+ countries - $R^2 = 0.977$ regression accuracy; automated anomaly detection using Isolation Forest across 3 research pipelines, surfacing data integrity issues that manual review missed.
- Designed Tableau, ggplot2, and Quarto dashboards for faculty leadership - **40% improvement in research comprehension** across two cohorts.

Magizhs Tech Solution - AI/ML Engineer Intern

May 2025 - Aug 2025

AI/ML Engineer Intern - Agentic Pipelines · RAG · Data Validation

- Built agentic mission-planning data pipeline (LangGraph stateful graph) with RAG retrieval over FAA Part 107 regulatory corpus - automated compliance data validation enforcing altitude, battery, and geometric constraints before export.

PROJECTS

Leasel: AI Document Intelligence & Evaluation Infrastructure

Data Infrastructure · LLM Evaluation

Built end-to-end AI data and evaluation infrastructure for affordable housing compliance: (1) **Data ingestion pipeline** - OCR-fallback ingestion (pdfplumber, pytesseract), section-aware chunking, TF-IDF retrieval over 1,240 residential lease PDFs across 5 US states; SQLite knowledge repository storing documents, chunks, settings, tickets, and release notes; (2) **Evaluation harness** - 225-query benchmark dataset, pass@k multi-trial sampling to detect non-deterministic variance, LLM-as-judge scoring across factual accuracy / grounding / citation correctness / hallucination, regression gate that blocked deploys on score drops - **93.6% precision, 87.9% F1, 0% unsupported-claim rate**; (3) 8-agent LangGraph pipeline (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → HITL Escalation) with live HUD income-limit API integration.

Python · LangGraph · TF-IDF · SQLite · FastAPI · OCR · LLM-as-Judge · Regression Gating

AgentProbe: AI Model Evaluation & Benchmarking Infrastructure

AI/ML Workflows · LLM Evaluation

Open-source AI evaluation and benchmarking toolkit: **5-metric eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance) integrated into CI/CD to gate production deployments; **LLM Router** for dynamic multi-provider routing (Anthropic, OpenAI, Groq, Ollama); 3 LangGraph agents (Orchestrator, RAG Tracer, Auto Fix) with per-run metric capture; 15-endpoint FastAPI backend; React observability dashboard with regression drift detection across evaluation runs. Runs 100% locally at zero cost.

Python · LangGraph · ChromaDB · FastAPI · React · CI/CD (GitHub Actions) · LLM-as-Judge

CrashClarity: Analytics Engineering on 104K Road Safety Records

Analytics Engineering

Built data pipeline and ML modeling workflow on 104K+ UK accident records - feature engineering across weather, road, and time variables; Random Forest + Logistic Regression achieving **85%+ accuracy**; deployed interactive analytics dashboard on GitHub Pages.

Python · scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages

EDUCATION

M.S. Data Analytics Engineering — George Mason University — May 2026

GPA: 3.87 | Virginia, USA

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University — 2017

TECHNICAL SKILLS

Data Engineering: Python, SQL, PL/SQL, Apache Spark, PySpark, ETL/ELT pipelines, data ingestion, data quality & integrity, data partitioning, performance optimization, schema design, data modeling, data orchestration, pipeline automation, pipeline monitoring

Databases: PostgreSQL, MySQL, Oracle, SQLite, RabbitMQ, NoSQL / non-relational databases

Cloud & Infrastructure: AWS S3, EC2, RDS, AWS Glue (familiar), Redshift (familiar), cloud-native data architectures, Docker, CI/CD (GitHub Actions), microservices, distributed systems

AI/ML Data Workflows: LLM evaluation infrastructure, LLM-as-judge, benchmark datasets, regression gating, RAG pipelines, LangGraph, LangChain, ChromaDB, FastAPI

ML & Analytics: scikit-learn, PyTorch, XGBoost, Random Forest, Isolation Forest, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Tableau, Power BI, ggplot2, Jupyter

Languages: Python, PHP, JavaScript, TypeScript, R, Bash, SQL/PL/SQL